

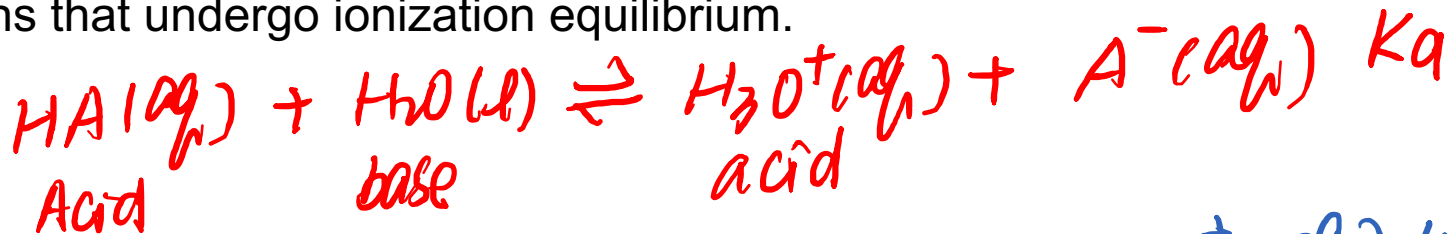
Final Exam Review

Dr. Wang

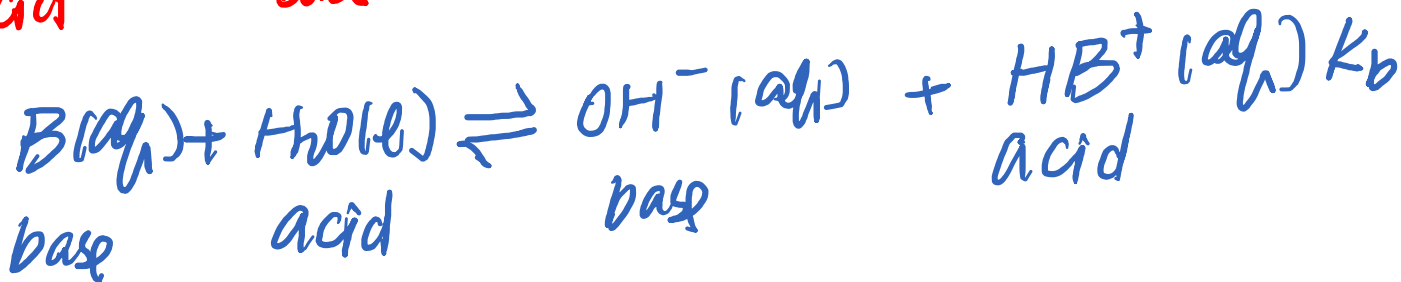
Unit 4

- Compounds/ions that undergo ionization equilibrium.

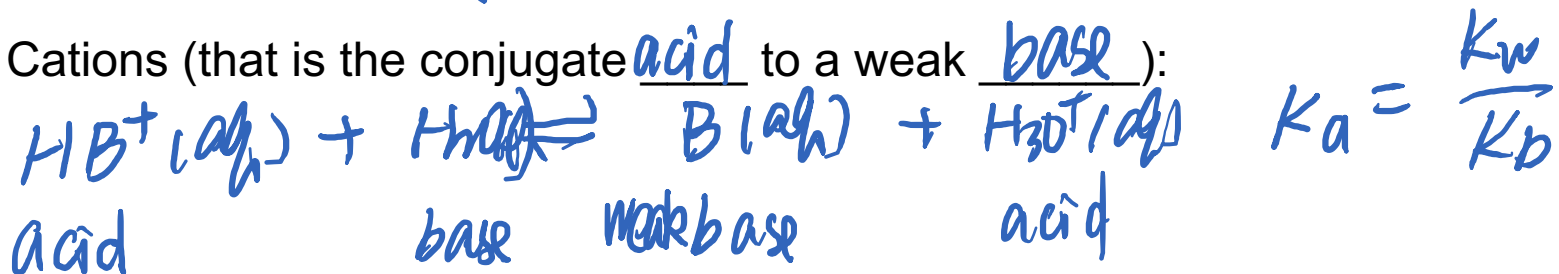
Weak Acid:



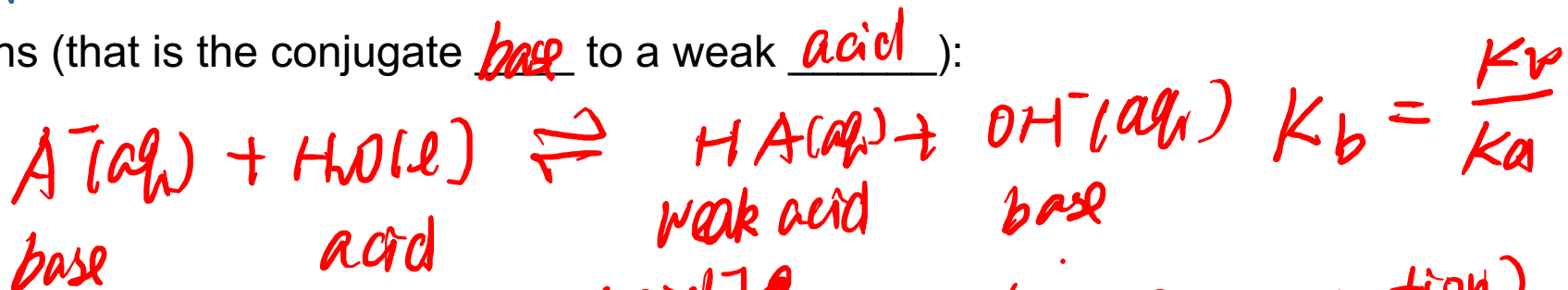
Weak Base:



Cations (that is the conjugate acid to a weak base):



Anions (that is the conjugate base to a weak acid):



- % ionization =

$$\frac{[\text{conjugate base/acid}]}{[\text{weak acid/base}]} \times 100\% < 5\% \text{ (estimation)}$$

Unit 4

pH of a salt solution

1. Dissociation: $AB(aq) \rightarrow A^+(aq) + B^-(aq)$

2. Ionization? (Look at their conjugate pairs!)

Cation	Anion	pH?
weak	strong	acidic / $pH < 7$
strong	strong	neutral / $pH = 7$
strong	weak	basic / $pH > 7$

Formula	K_a
HIO ₃	1.7×10^{-1}
HC ₂ H ₃ O ₂ Cl	1.4×10^{-3}
HNO ₂	4.6×10^{-4}
HF	3.5×10^{-4}

Formula	K_b
C ₄ H ₉ NH ₂	5.9×10^{-4}
CH ₃ NH ₂	4.5×10^{-4}
NH ₃	1.8×10^{-5}

Which one of the following salts, when dissolved in water, produces the solution with the **highest** pH?

A) NaNO₂

B) KClO₄

C) NH₄Cl

D) LiF

NO_2^-

$\leftrightarrow$

HNO₂

basic

ClO_4^-

$\leftrightarrow$

HClO₄

neutral

Cl^-

$\leftrightarrow$

HCl

acidic

F^-

$\leftrightarrow$

HF

basic

strongest

weakest

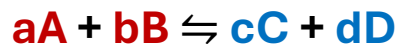


Announcements 5/15

Final Exam at 1:30 PM-4:00PM on Mon 5/18

- Learning objective uploaded: Unit 1-3: 40%; Unit 4: 60%
 - Calculators (any kind), pens/pencils/highlighters, and erasers are allowed.
 - Bonus point for Unit 4: Chem RePS worksheet and Exam 3 Wrapper
- 3rd
- Office hour
Dr. Wang 5/15 Friday 1:00 - 3:00 PM at KC 278
 - SI Session
Paige Kwan 5/16 Saturday 11:00 AM- 1:00 PM at BK 209

Summary on Le Châtelier's Principle



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

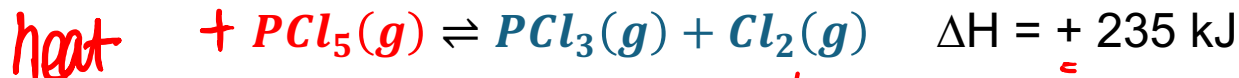
endo: heat $\rightarrow$
exo: $\rightarrow$ heat

New equilibrium shift to the direction to reduce the introduced changes...

Change	Shift Equilibrium?	If So, How?	Change K ?
(g) (aq) Concentration	<u>yes</u>	increase [reactant], shift to <u>right</u> decrease [product], shift to <u>right</u> increase [product], shift to <u>left</u> decrease [reactant], shift to <u>left</u>	<u>no</u>
Volume / Pressure	yes, but only when $\Delta n \neq 0$ (g) (aq)	P $\uparrow$ or V $\downarrow$, shift to <u>less</u> moles of gases P $\uparrow$ at fixed V, no effect P $\downarrow$ or V $\uparrow$, shift to <u>more</u> moles of gases	<u>no</u>
Temperature	<u>yes</u>	T $\uparrow$, shift to the direction <u>consume</u> heat T $\downarrow$, shift to the direction <u>produce</u> heat	$\Delta G^\circ = -RT \ln K$ <u>yes</u>
Catalysts	<u>no</u>		<u>no</u>

Your turn!

Consider the following reaction:



Which disturbance will lead the equilibrium shift to the right?

A. Remove Cl_2 $\rightarrow$
right

B. Decrease the total pressure of the reaction
 $P \downarrow, V \uparrow \Rightarrow$ diluted $\Rightarrow$ more (g) $\Rightarrow$ shift to right $1 \rightarrow 2$

C. Decrease the temperature of the reaction

$\leftarrow$ shift left $Q < K = \frac{[\text{P}]}{[\text{R}]}$

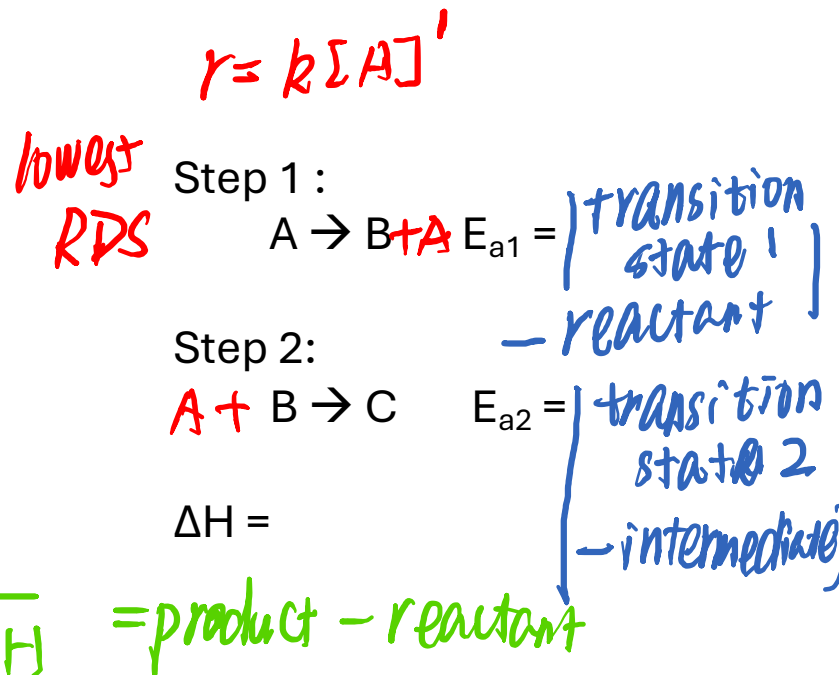
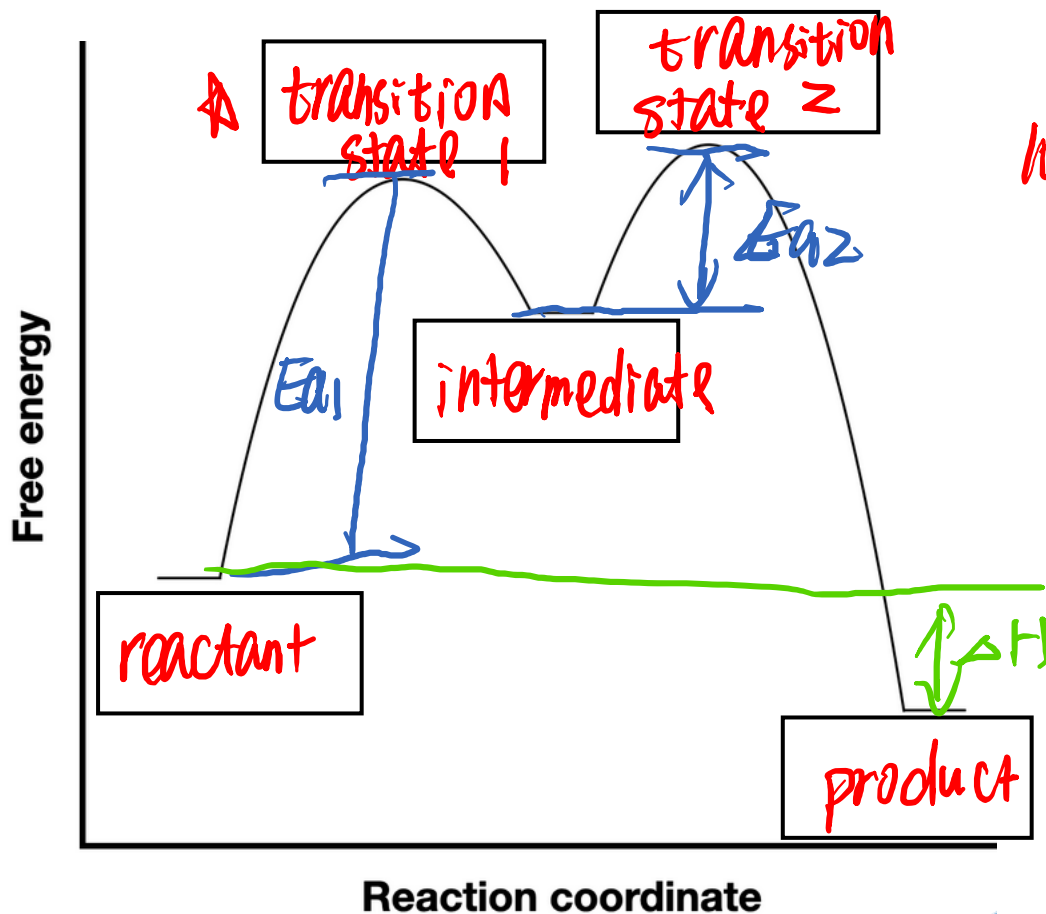
$Q > K$	Backward
$Q = K$	at equilibrium
$Q < K$	forward

$K > 1$	favor product
$K = 1$	
$K < 1$	favor reactant



Unit 3

1. Interpret a Potential Energy (Reaction Coordinate Diagram) to determine the energy of activation for a reaction and whether the reaction is Exothermic or Endothermic.

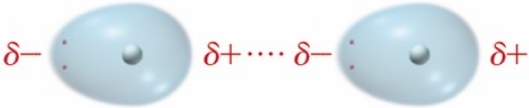
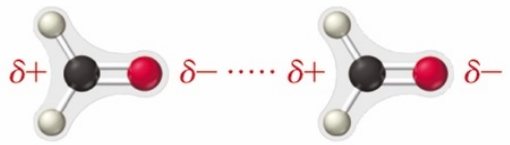
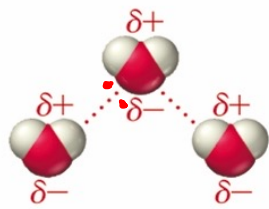
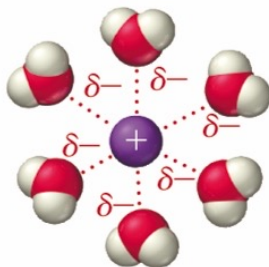


$$k = A e^{-\frac{E_a}{RT}}$$

rate constant

Unit 2

- Determine types and relative strengths of intermolecular forces present in a sample based on the compound's structural formula

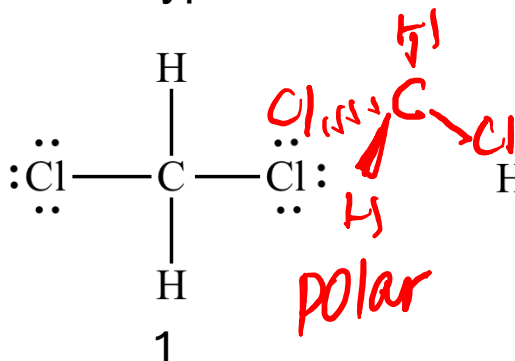
Type	Present In	Molecular Perspective	Strength
Dispersion	all		0.05–20+ kJ/mol
D – D	polar		3–20+ kJ/mol
H-bond	have H N/O/F – H N/O/F		10–40 kJ/mol
ion – dipole (polar)	solution		30–100+ kJ/mol

*The dispersion force can become very strong (as strong and even stronger than the others) for molecules of high molar mass.

Unit 2

1. Determine types and relative strengths of intermolecular forces present in a sample based on the compound's structural formula
2. Relate miscibility of liquids to structure of liquids

What types of intermolecular forces do these compounds have?

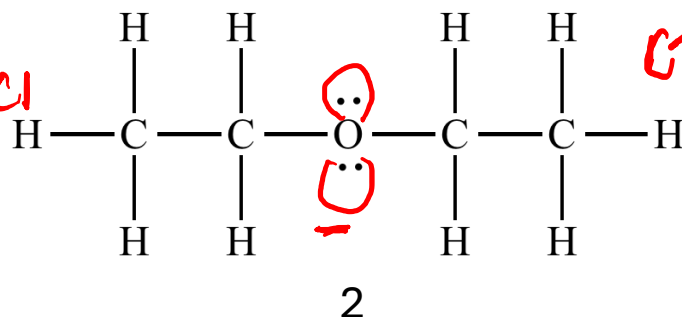


Dispersion? ☒

Dipole-dipole? ☒

H bonding? ☒

Ion-dipole? ☒

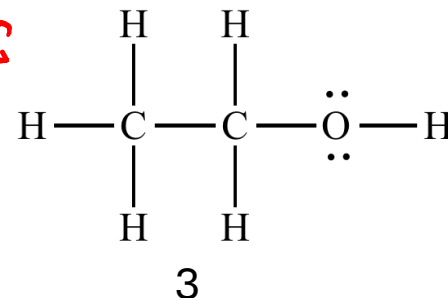


Dispersion? ☒

Dipole-dipole? ☒

H bonding? ☒

Ion-dipole? ☒



Dispersion? ☒

Dipole-dipole? ☒

H bonding? ☒

Ion-dipole? ☒

Are they miscible? ☒

polar

dissolve

polar

/

nonpolar

dissolve

nonpolar

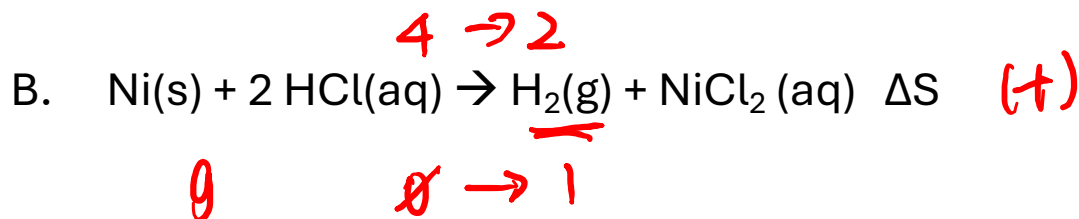


Unit 1

1. Understand the concept of spontaneous change and second law of thermodynamics (Entropy of the Universe)
2. Understand the concept of entropy and predict the sign of entropy.
3. Predict the sign of entropy of the system/entropy of the surroundings for a process or chemical reaction.

Entropy (S) increases when ...

- The volume ↑
- The temperature (average kinetic energy of particles) ↑
- the number of molecules ↑
- Physical state changes: S(gas) ≫ S(liquid) > S(solids)
- S (pure substances) < S(mixtures/solutions)
- the complexity of molecule ↑



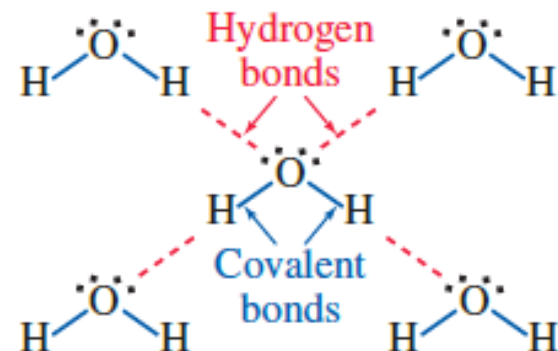
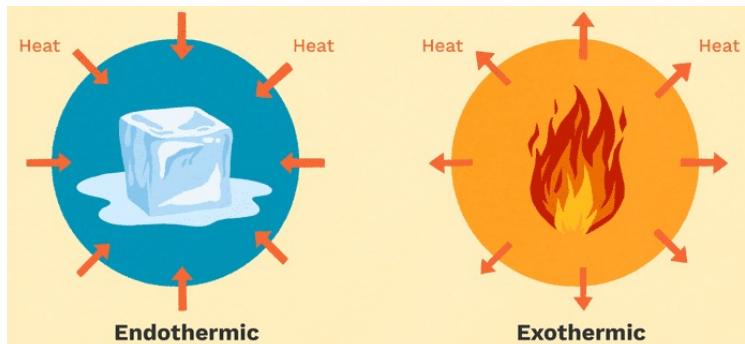
Unit 1

1. Predict spontaneity from ΔH and ΔS .

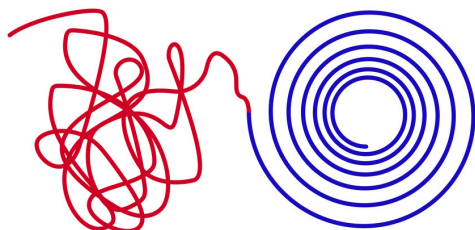
$$\Delta G = \Delta H - T\Delta S$$

$\left\{ \begin{array}{l} (-) \text{ spontaneous} \\ (+) \text{ non spontaneous} \end{array} \right.$

ΔH	ΔS	ΔG	Spontaneous?
(+)	(-)	(+)	non spontaneous
(+)	(+)	high T	spontaneous
(-)	(-)	low T	spontaneous
(-)	(+)	(-)	spontaneous



Chaos → Order



We make it!!
Thanks for the
fantastic year!

